NEWSBOT INTELLIGENCE SYSTEM 2.0

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ITAI 2373 – NLP Final
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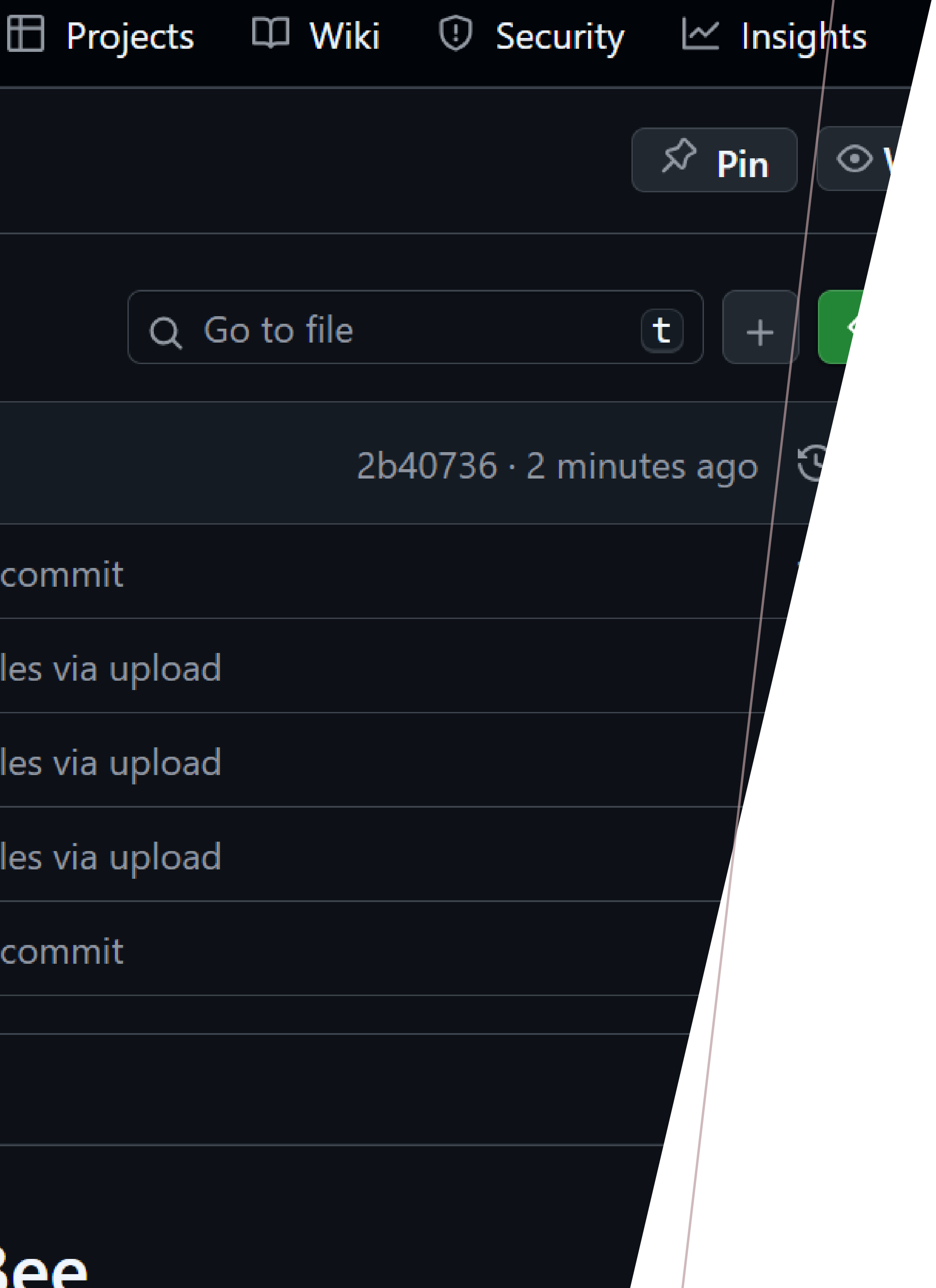

AGENDA

Hi, I'm Bryan Tzorin. This is my final project. NewsBot 2.0 is an NLP system for news from classification to topics, POS & sentiment, summaries, semantic searches, multilingual news and an interactive demo as well. Runs all in Colab and saves outputs for grading.

GOALS & WHAT IT DOES

- CLEAN, CONSISTENT PIPELINE THAT RUNS IN COLAB
- CLASSIFIER + TOPICS + POS + SENTIMENT
- SUMMARIES + SEMANTIC SEARCH
- MULTILINGUAL PASS (TRANSLATE → ANALYZE)
- SIMPLE INTERACTIVE DEMO (GRADIO)

The system does more than assign a predicted category. It finds themes, generates article abstracts, gauges sentiment, and searches for comparable pieces. Everything is interconnected so it can be assessed and repurposed.

DATASET & DATAPREP

The CSV didn't always have labels. In order to keep the supervised pipeline intact, I created labels using clustering and made small clusters into "misc." This avoids stratified split errors and allows access to categories down the line.

- BBC-style news CSV (sometimes no labels)
- Light cleaning → bbc_clean.csv
- Auto-label via TF-IDF + MiniBatchKMeans (if labels missing)
- Stratified 80/20 split

Classification

- TF-IDF (1–2 GRAMS, ~50K FEATURES)
- MODELS: LOGISTIC REGRESSION & LINEAR SVM
- AUTO-SELECT BEST AND EXPORT METRICS.JSON
- CONFUSION MATRIX + MISCLASSIFIED SAMPLES

Linear baselines work well for sparse text - such as this one. I preserved accuracy, per-class metrics, and a confusion matrix, along with a table of misclassifications for error analysis.

TOPICS (NMF + LDA)

NMF YIELDS A CLEARER TOP-TERM SETS; LDA REVEALS THE PROBABILITIES OF THE TOPICS. BOTH HELP TO DEFINE THE CORPUS AND BRING CONTEXT BEYOND THE CLASSIFIER.

- NMF on TF-IDF → interpretable topics
- LDA on bag-of-words → probabilistic view
- topics_overview.csv & doc_topics_nmf.csv

POS & SENTIMENT

WHERE SPORTS RECAPS TEND TO HAVE MORE VERBS, TECH/BUSINESS PIECES ARE MORE NOUN HEAVY. THUS, VADER IS VERY QUICK AND EFFICIENT FOR GENERAL ENGLISH, MOSTLY LIKE A HEADLINE.

- POS (NLTK universal tagset) → style differences across categories
- VADER sentiment (neg/neu/pos + compound)
- Aggregate CSVs saved for grading

SUMMARIES & SEARCH

Summaries transform lengthy pieces into something scannable. Search finds pertinent contextual information in a flash. These two features are incredibly helpful to real users to quickly assess and explore narratives of their interest.

- Extractive summaries (TextRank via sumy) → summaries.csv
- TF-IDF cosine search → search_examples.csv
- Example queries across sports/politics/tech

MULTILINGUAL PASS

I FEEL THAT THIS IS A VERY REALISTIC SOLUTION FOR COLAB AS WELL, TRANSLATE AND USE THE ENGLISH PIPELINE AGAIN. NOT ENTIRELY IDEAL BUT IT SHOWS THE CROSS-LANGUAGE PROCESSING NICELY.

- Translate non-English → English (deep-translator)
- Analyze with English tools (sentiment, classification optional)
- multilingual_eval.csv

INTERACTIVE DEMO (GRADIO)

It makes it a usable application and not just a bunch of code. It's good for testing and displaying everything in one place.

- Paste text → predict + sentiment + nearest documents
- Optional translate-first checkbox
- Great for demos and sanity checks

RESULTS, LIMITS, NEXT STEPS

WHEN IT COMES TO A COLAB-BASED FINAL, THE OPTIONS ARE BETWEEN RELIABILITY AND CLARITY. WHAT WOULD MAKE IT NEWSROOM READY? ENHANCED SEARCHING, A MINUSCULE ABSTRACTIVE SUMMARIZER, AND ENTITY ANALYSIS.

- Strong, reproducible baseline with clear artifacts
- Limits: TF-IDF search, extractive-only summaries
- Next: sentence-BERT search, small abstractive model, NER/linking

THANK YOU

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